Real-Time Document Scanner and Chromatic Verification Mark Extractor Using Planar Homography and Adaptive Image Segmentation

Definition

A real-time Computer Vision application that automatically detects, rectifies, and digitizes physical documents from live camera streams by identifying quadrilateral paper boundaries via edge and contour approximation, eliminating perspective distortion through a 2D planar homography transformation, and applying parallel segmentation pipelines—adaptive binarization for shadow-free document scanning and HSV chromatic filtering for isolated extraction of pen signatures and official ink stamps.

Abstract

This project presents an efficient, real-time Computer Vision document digitization and verification pipeline executing on live webcam video streams at 30+ FPS. The incoming video frames are spatially normalized and smoothed using Gaussian and bilateral filtering to eliminate sensor grain while preserving critical structural edges. A hybrid edge-and-contour extraction module leverages Canny gradient tracking and Douglas-Peucker polygon approximation to continuously detect and stabilize the four corner coordinates of a target document in physical space, remaining robust against partial hand occlusions and ambient background clutter.

Perspective distortion caused by arbitrary camera tilt is dynamically corrected by calculating a 3x3 projective homography transformation matrix that warps the detected quadrilateral into an upright, canonical Euclidean rectangular plane. The rectified document is subsequently routed through dual-branch image processing pipelines:

• **Adaptive Spatial Thresholding:** Computes localized pixel-neighborhood thresholds to eliminate non-uniform lighting, hand shadows, and paper creases, generating a high-contrast digital scan.

• **Chromatic Segmentation:** Operates in the HSV color space utilizing selective hue-saturation bandpass filters to isolate handwritten blue/red pen ink and official validation stamps from printed text.

The system displays processing states through an interactive, multi-viewport diagnostic HUD displaying the live boundary tracker, Canny edge map, rectified document, and isolated verification marks, providing production-grade document scanning capabilities using deterministic classical algorithms without requiring deep learning models or specialized hardware.